

Mechanistic Consistency Across Five Mechanistically Diverse Enzymes

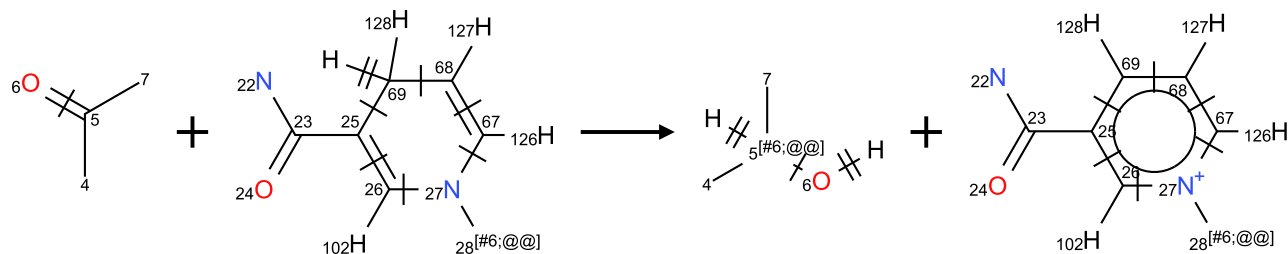
This file summarizes reaction-level mechanistic variation within five enzymes, each defined by a unique combination of organism, UniProt accession number, and EC classification. The enzymes were selected as those with the highest number of reactions, limited to entries with accession numbers and uniqueness at EC level 3. For each system, EVODEX-derived Bm, Cm, and Em operators were computed and aggregated across all associated reactions.

[1] Homo sapiens - EC 1.1.1.357

Protein: P52895 (uniprot) | Reactions: 128

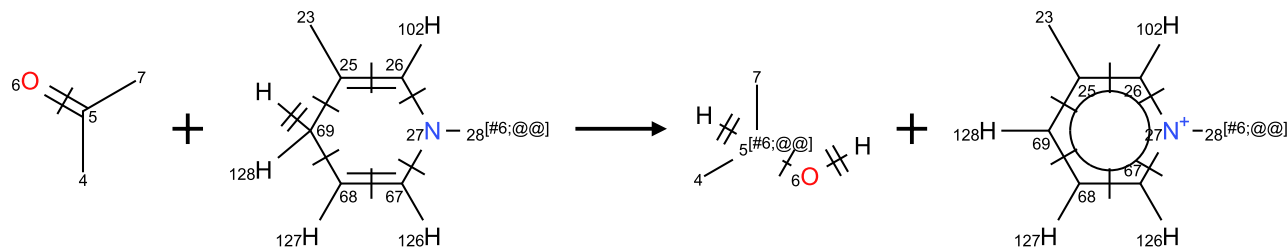
EVODEX-Em (1 unique)

(n=128)



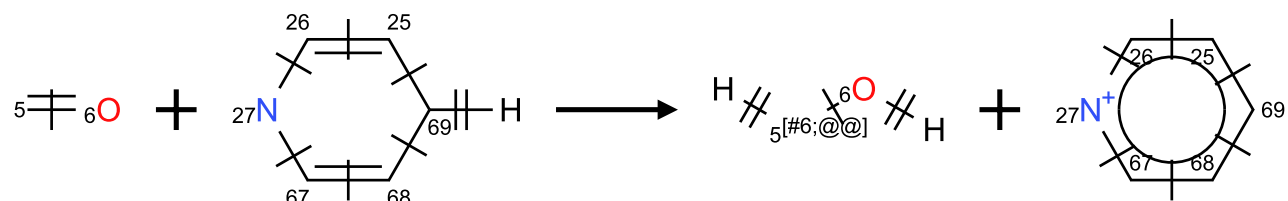
EVODEX-Cm (1 unique)

(n=128)



EVODEX-Bm (1 unique)

(n=128)

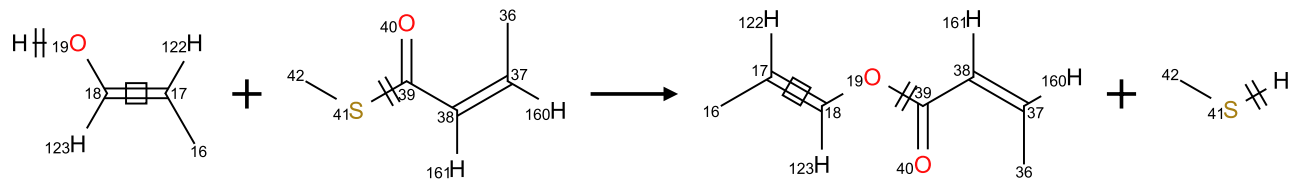


[2] *Marinobacter nauticus* - EC 2.3.1.75

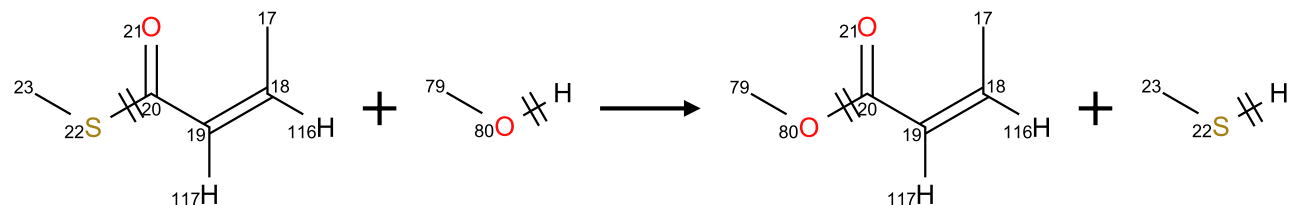
Protein: A3RE51 (uniprot) | Reactions: 190

EVODEX-Em (4 unique)

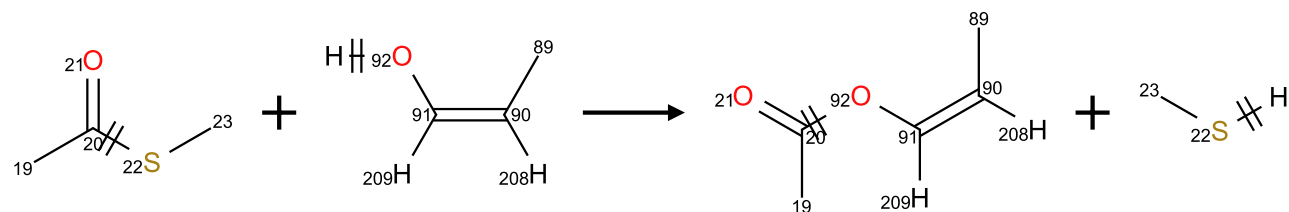
(n=124)



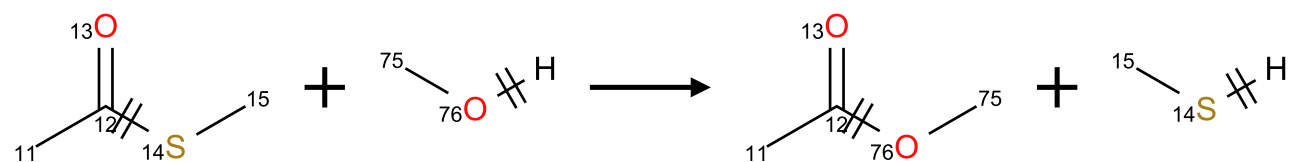
(n=49)



(n=11)

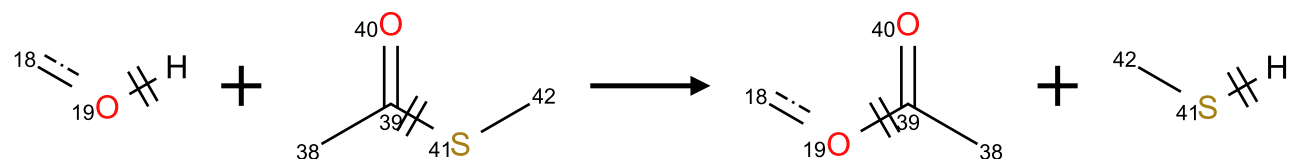


(n=6)



EVODEX-Cm (1 unique)

(n=190)



EVODEX-Bm (1 unique)

(n=190)

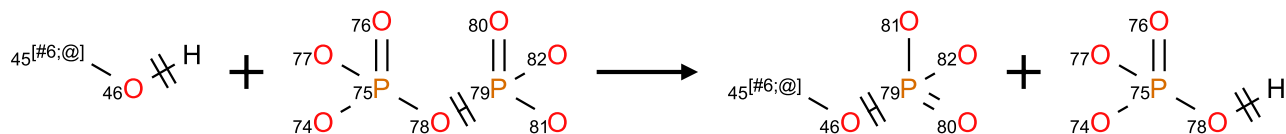


[3] *Listeria monocytogenes* - EC 2.7.9.6

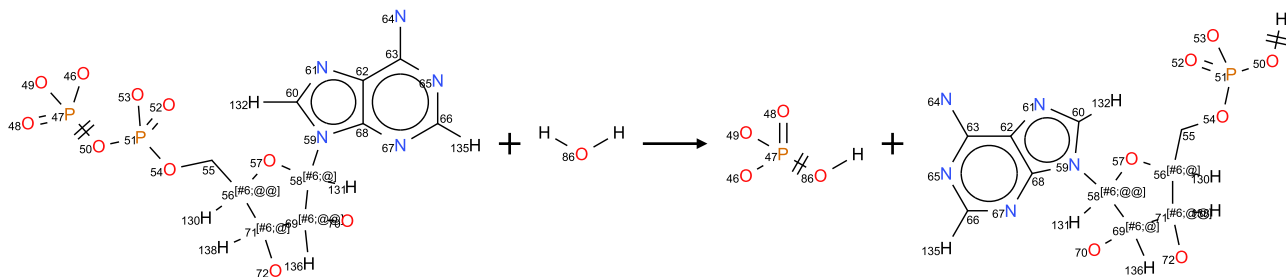
Protein: Q9RLT9 (uniprot) | Reactions: 154

EVODEX-Em (5 unique)

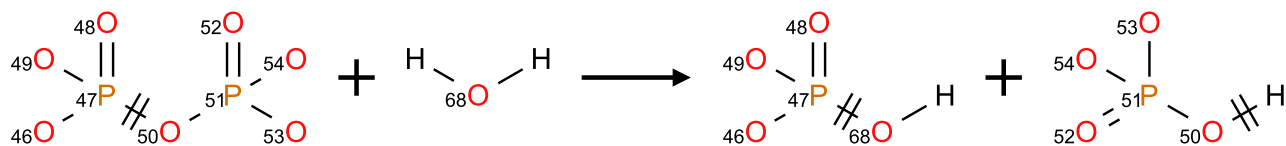
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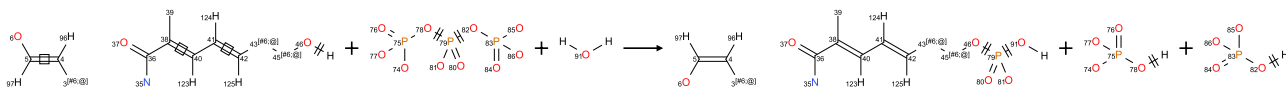
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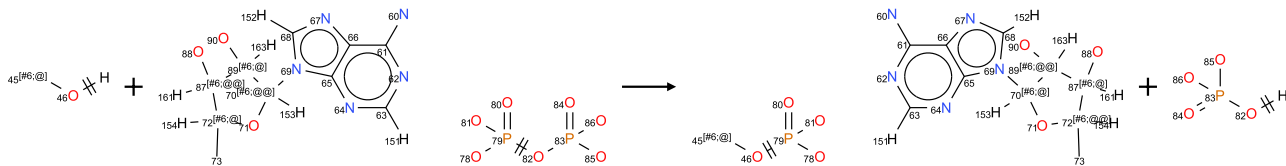
(n=14)



(n=14)

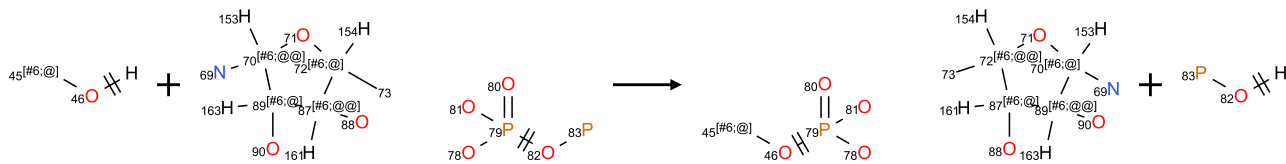


(n=14)

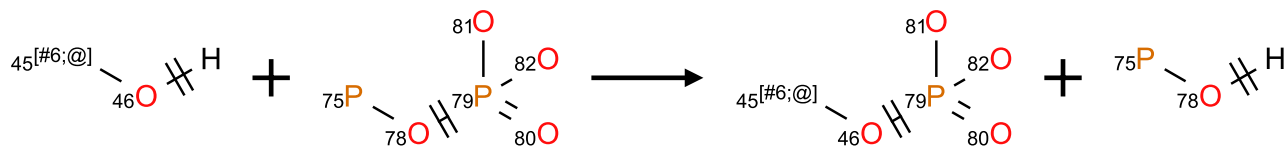


EVODEX-Cm (5 unique)

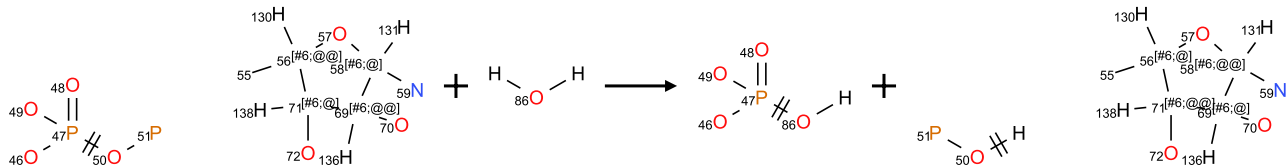
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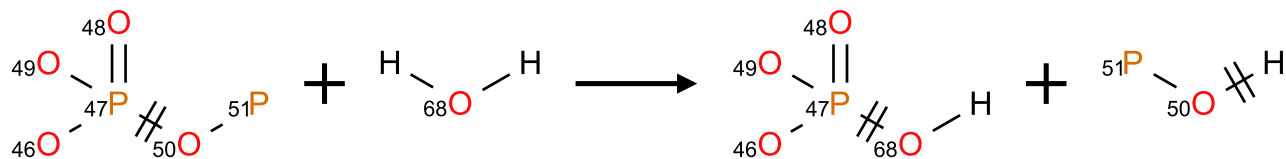
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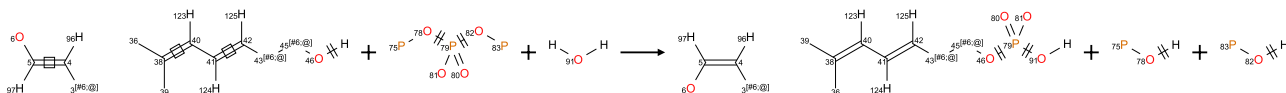
(n=14)



(n=14)



(n=14)

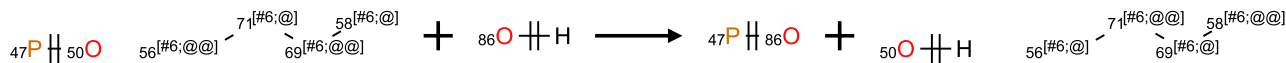


EVODEX-Bm (4 unique)

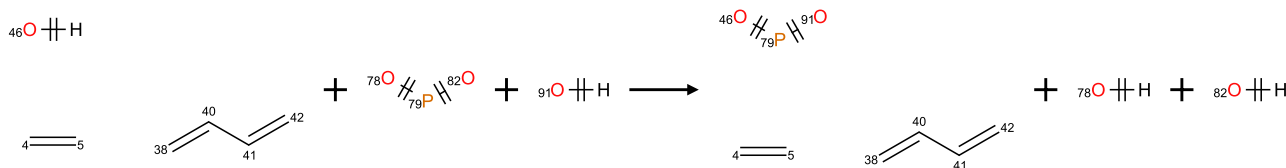
(n=112)



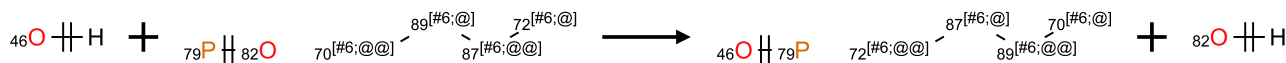
(n=14)



(n=14)

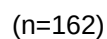
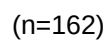


(n=14)



Protein: B5ARZ9 (uniprot) | Reactions: 164

(n=162)

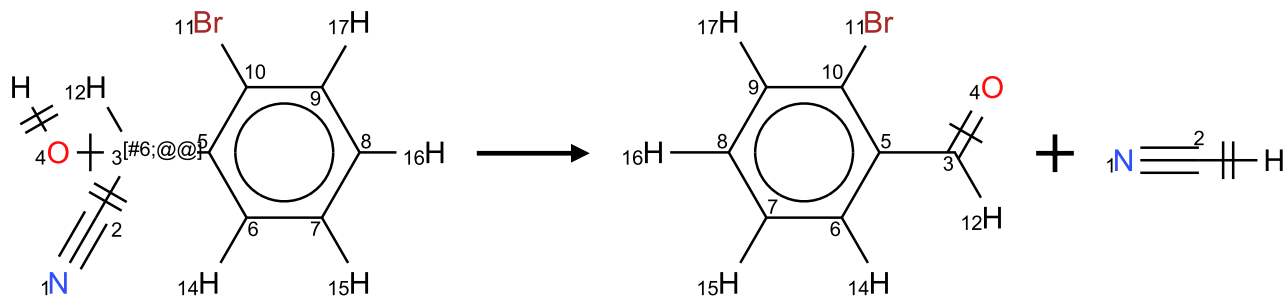


[5] Manihot esculenta - EC 4.1.2.47

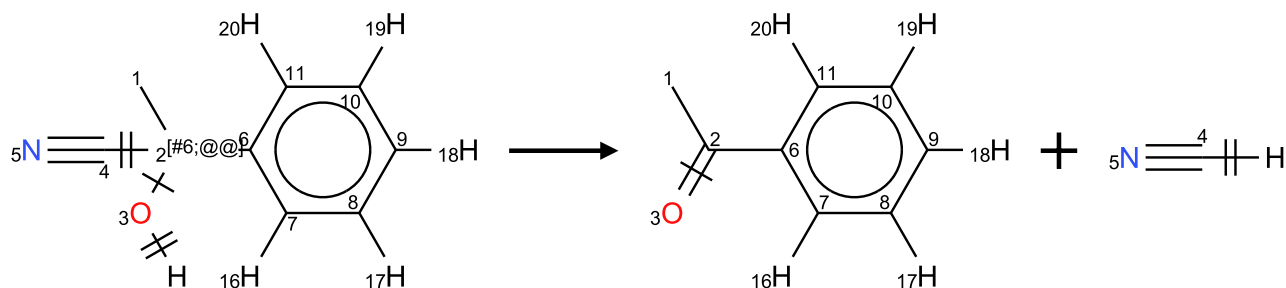
Protein: P52705 (uniprot) | Reactions: 138

EVODEX-Em (48 unique)

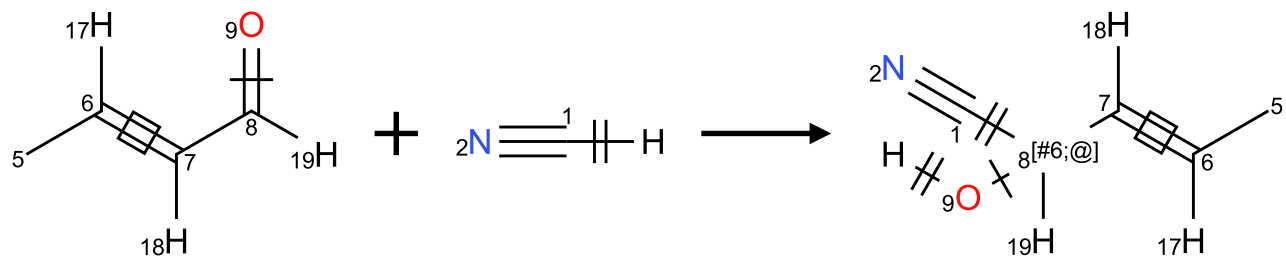
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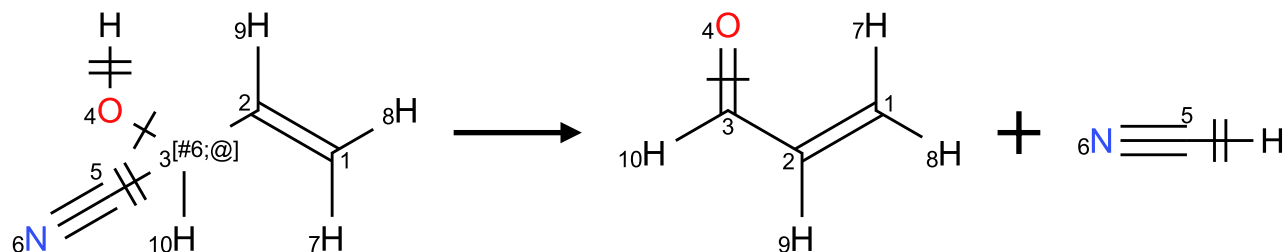
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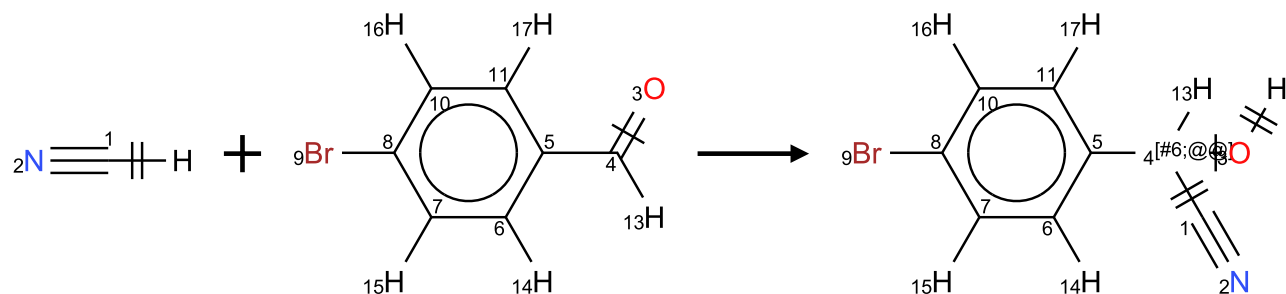
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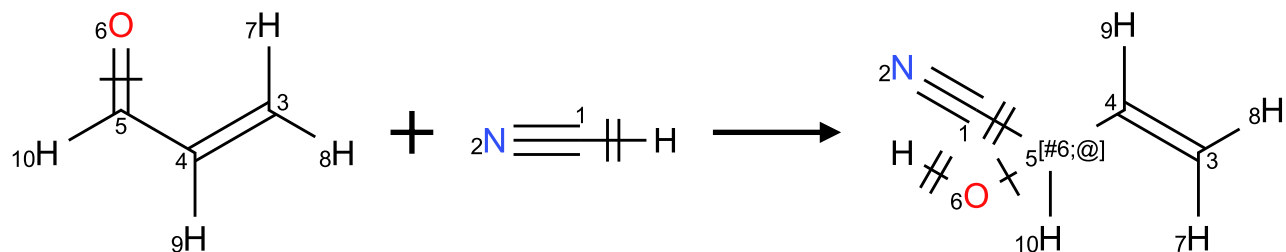
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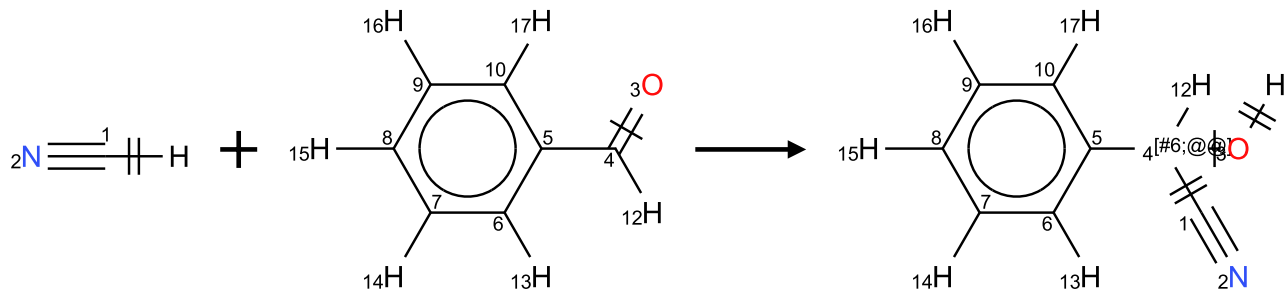
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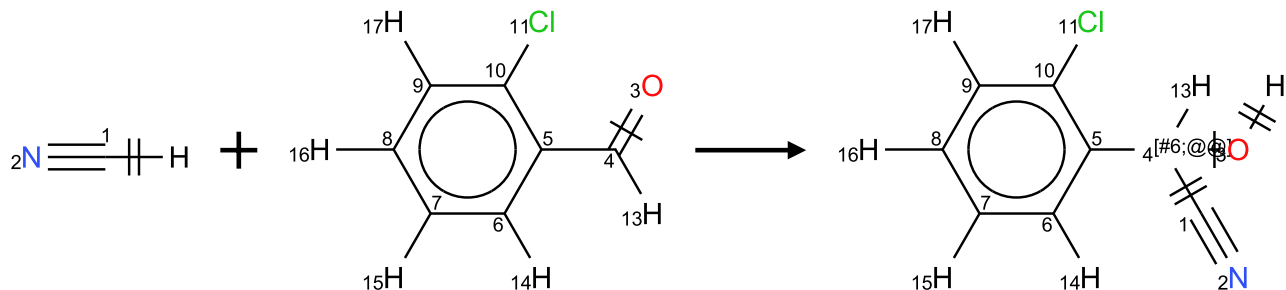
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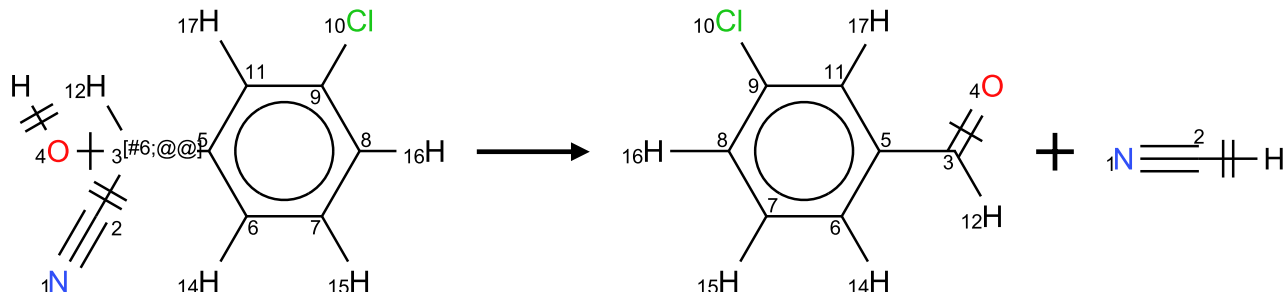
(n=2)



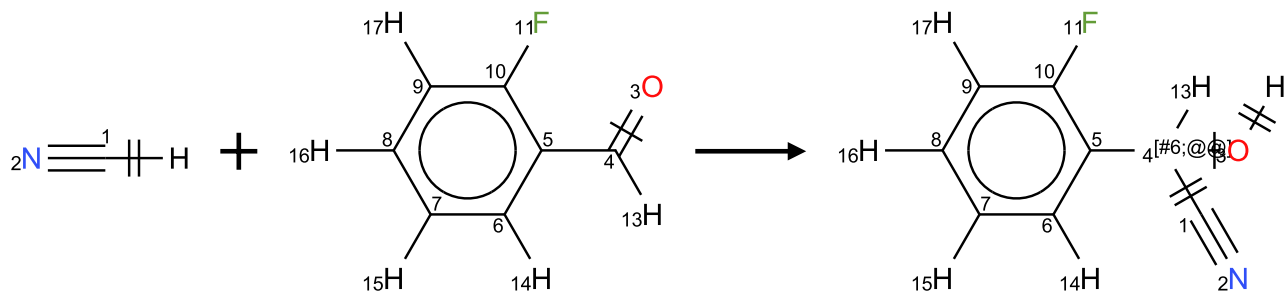
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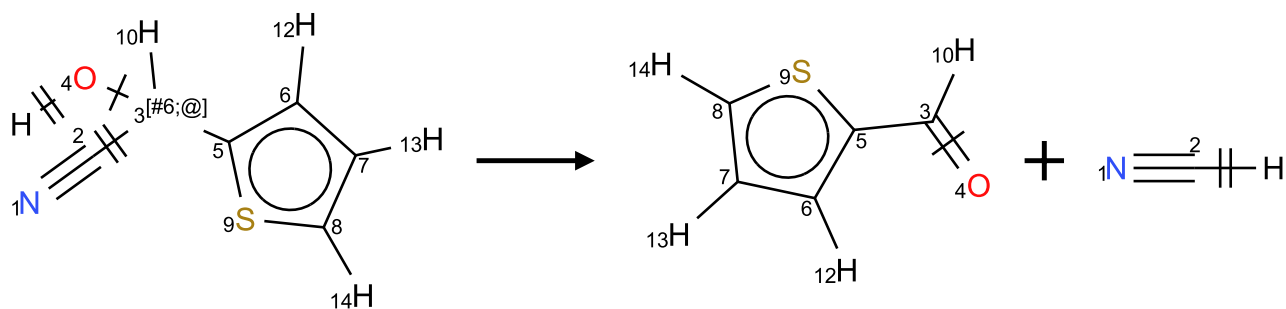
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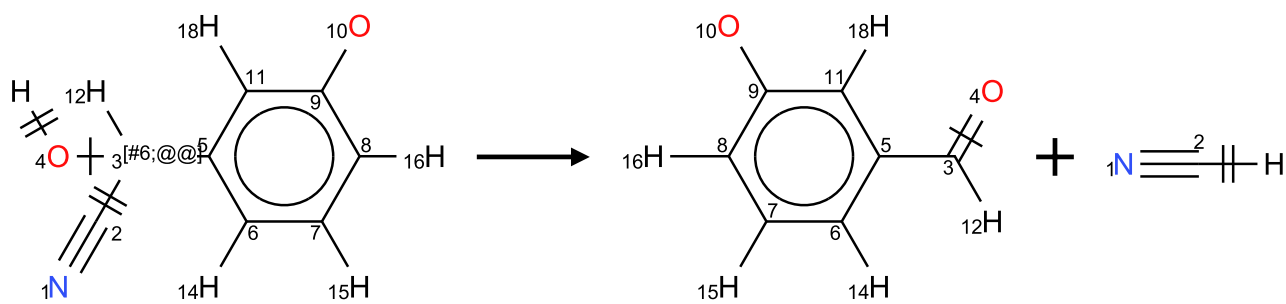
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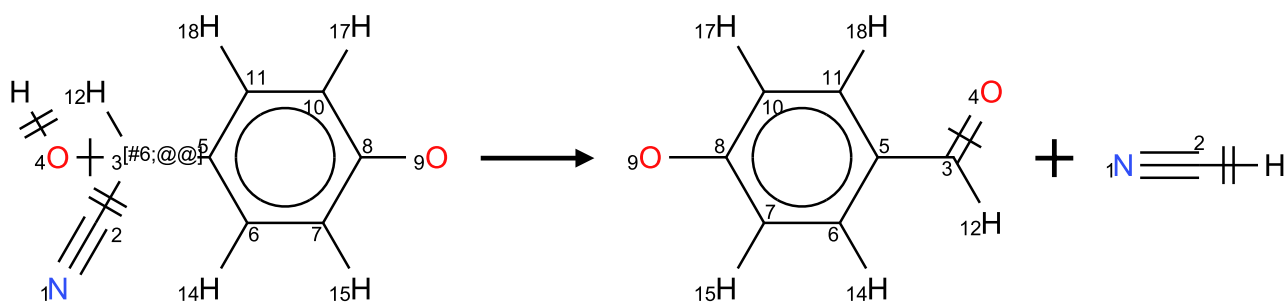
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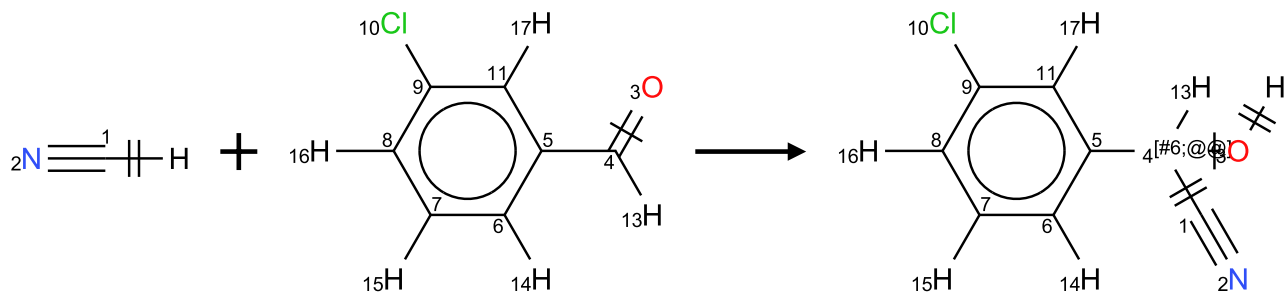
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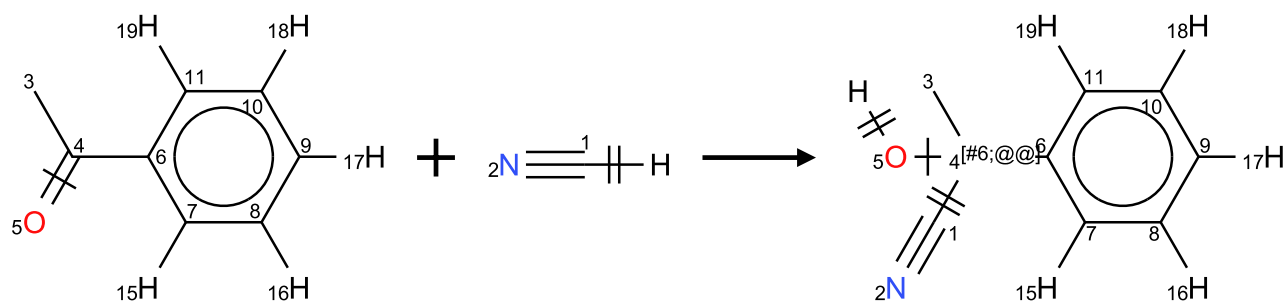
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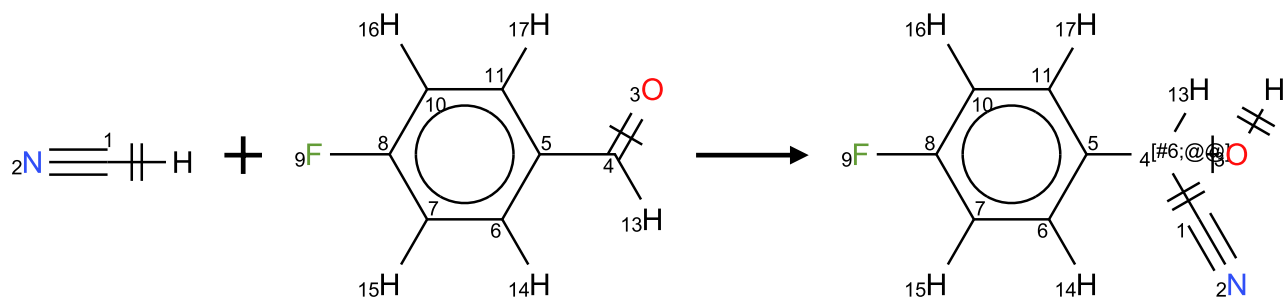
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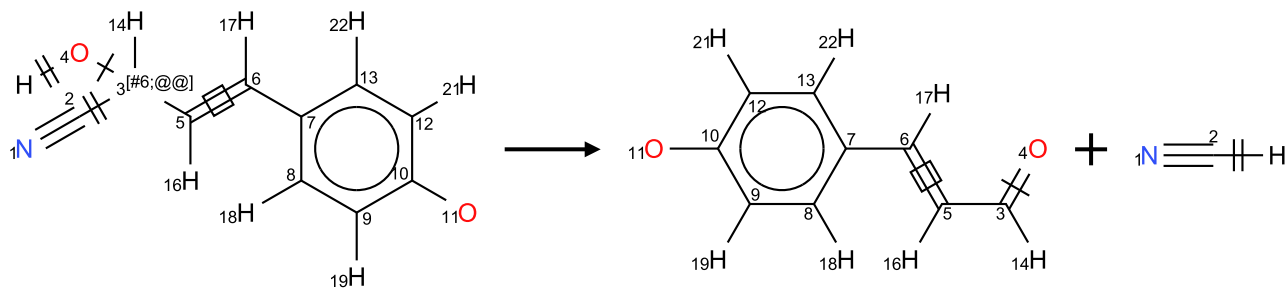
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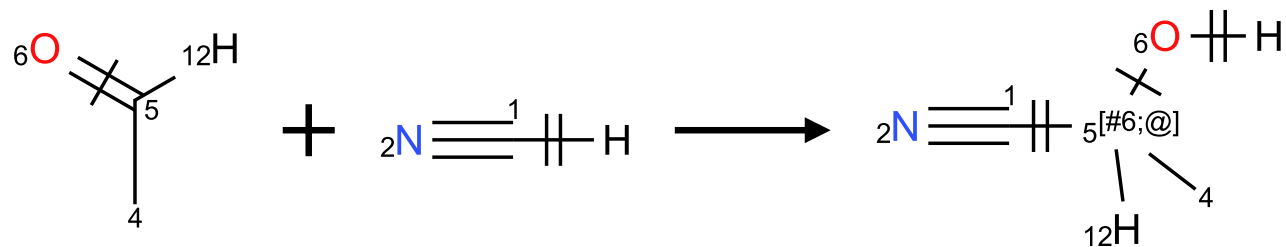
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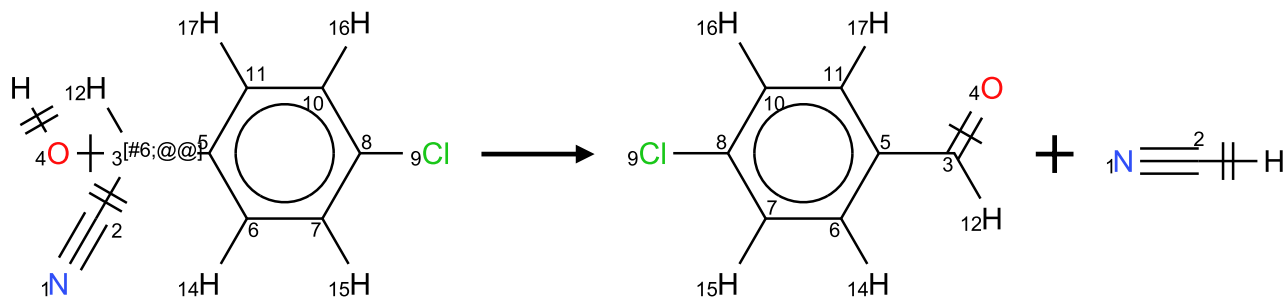
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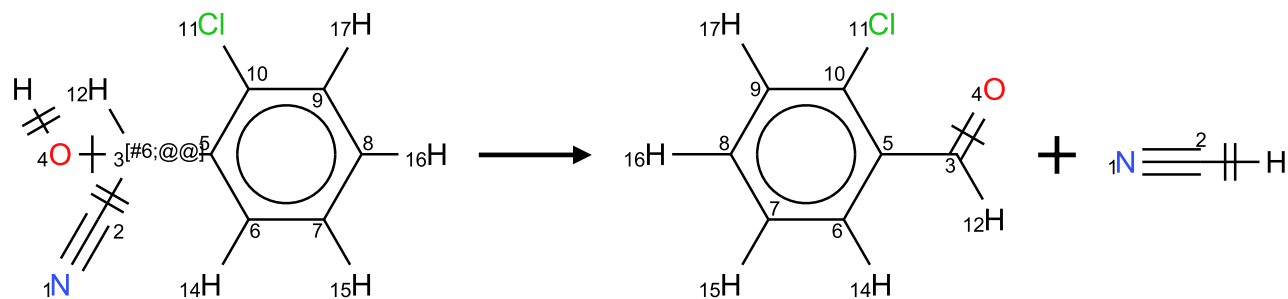
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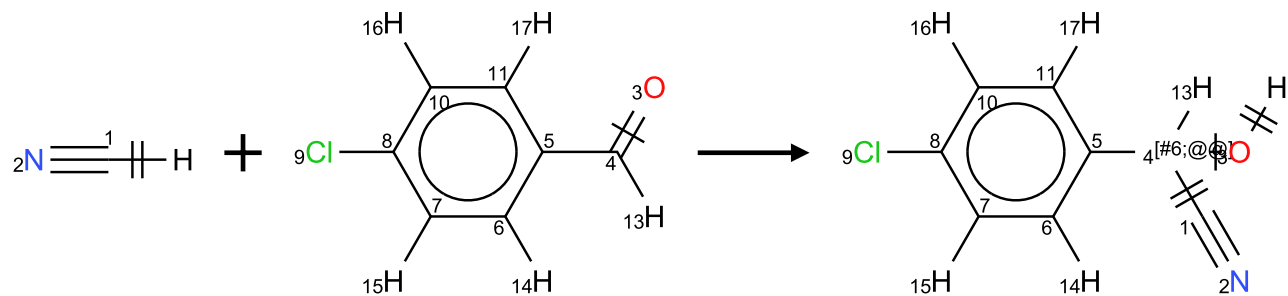
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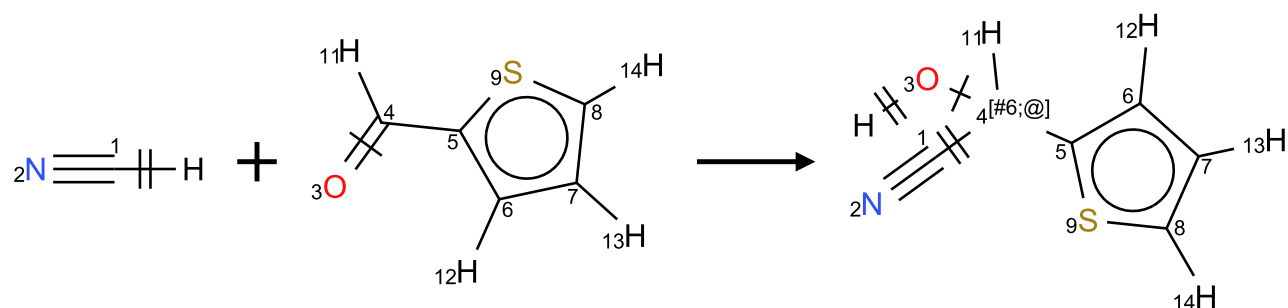
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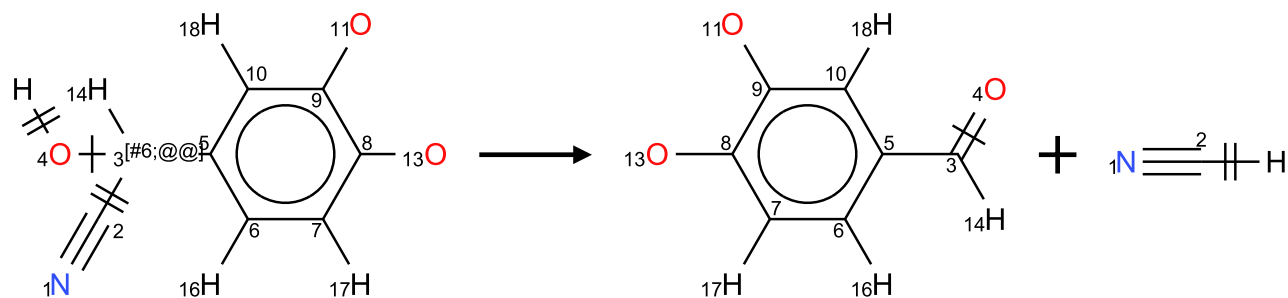
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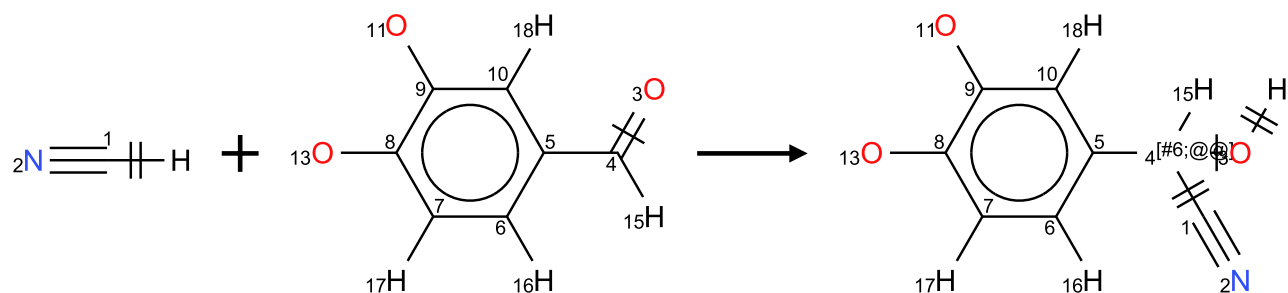
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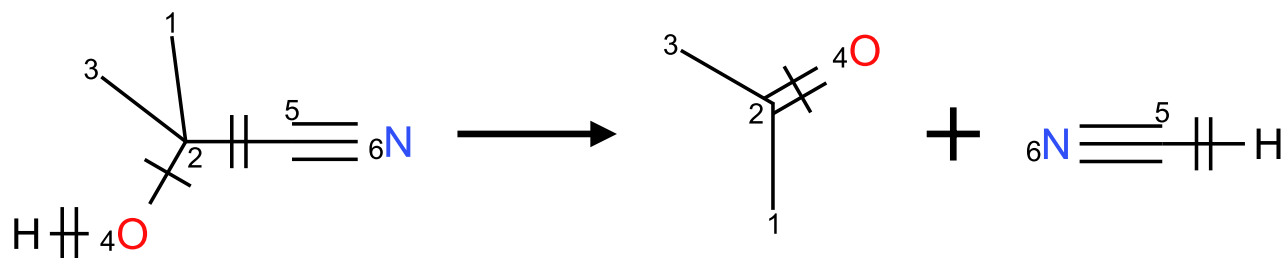
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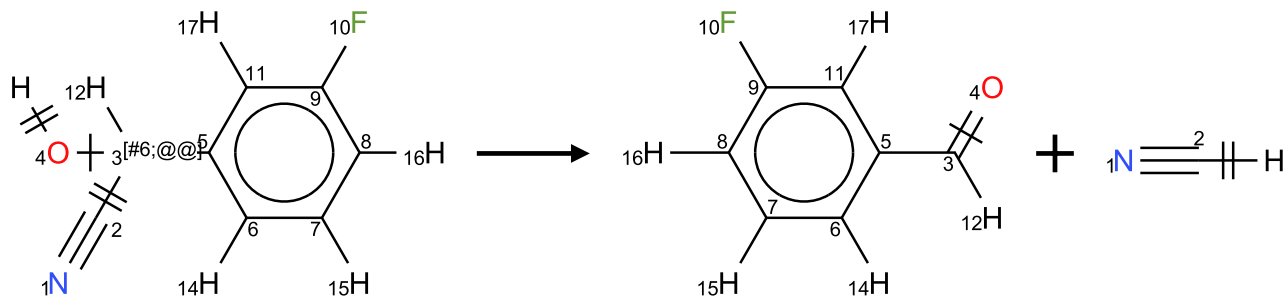
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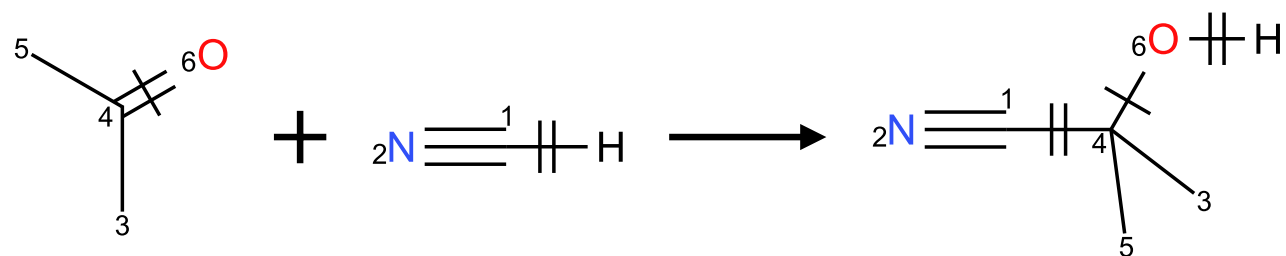
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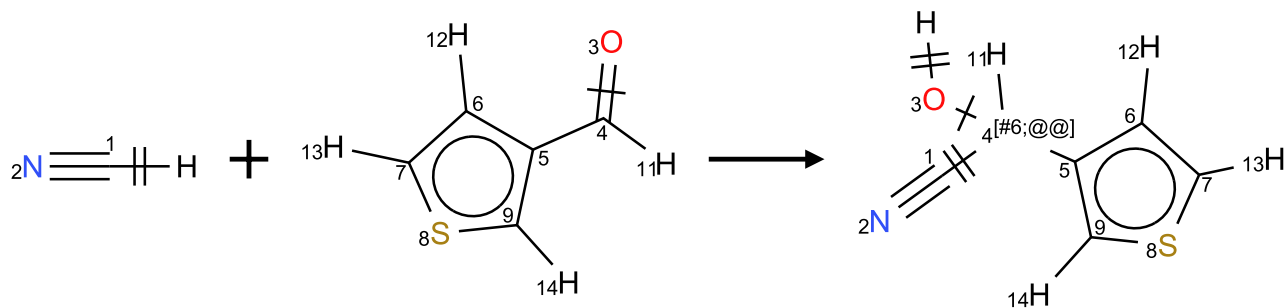
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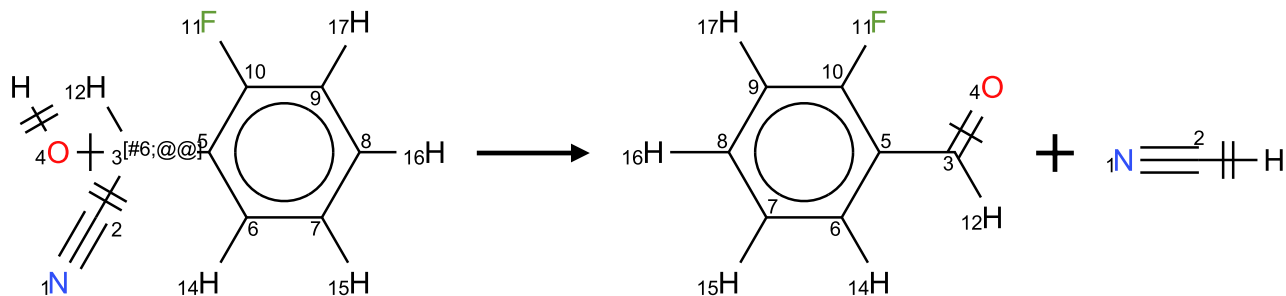
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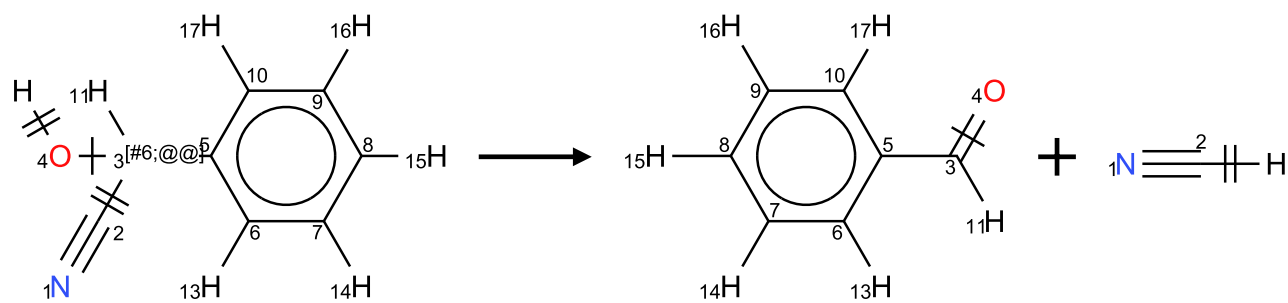
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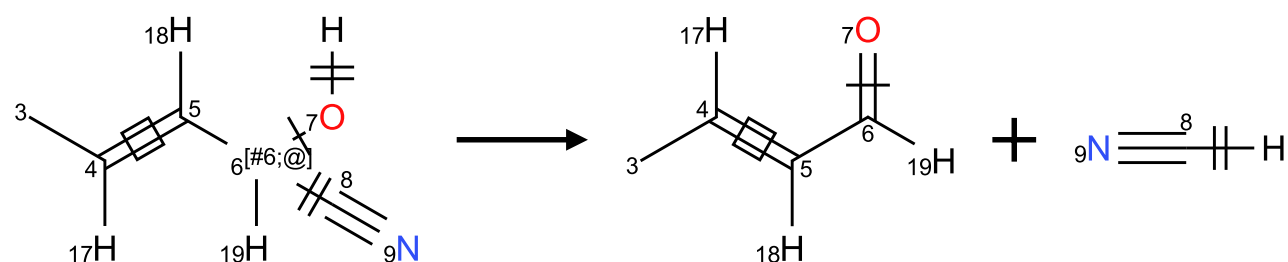
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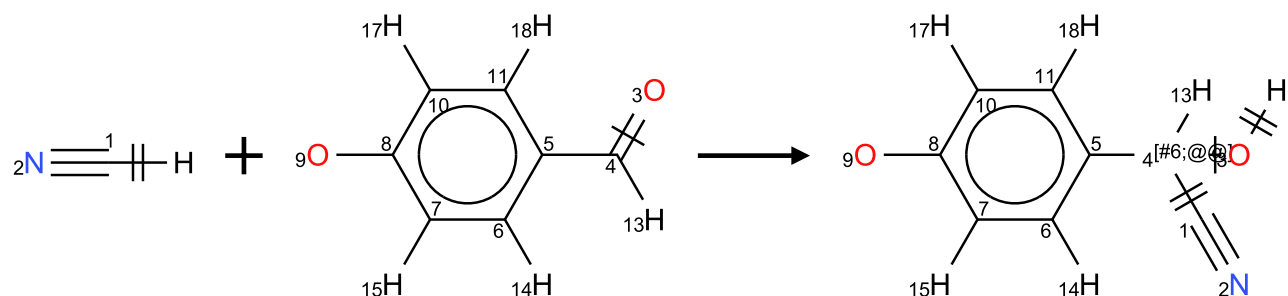
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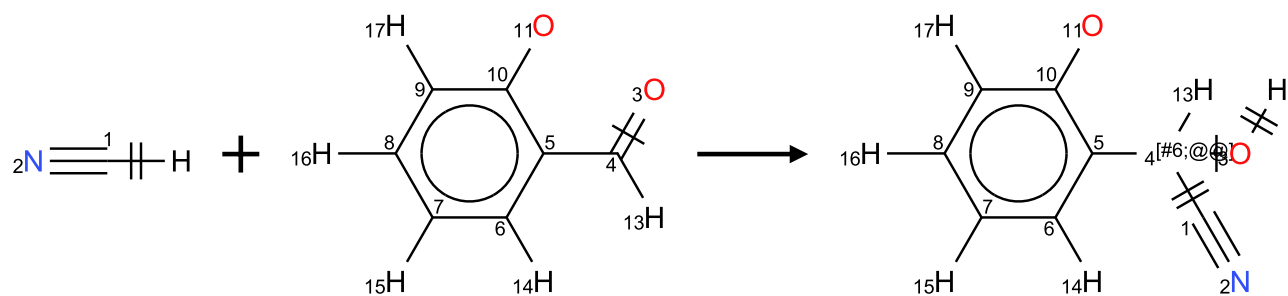
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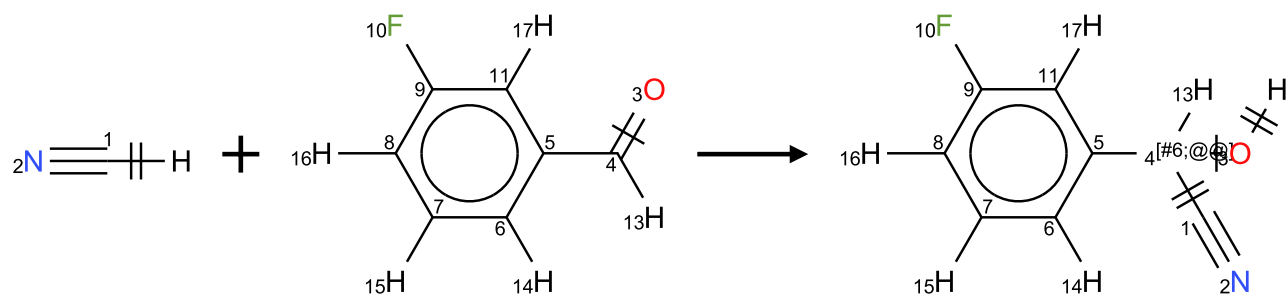
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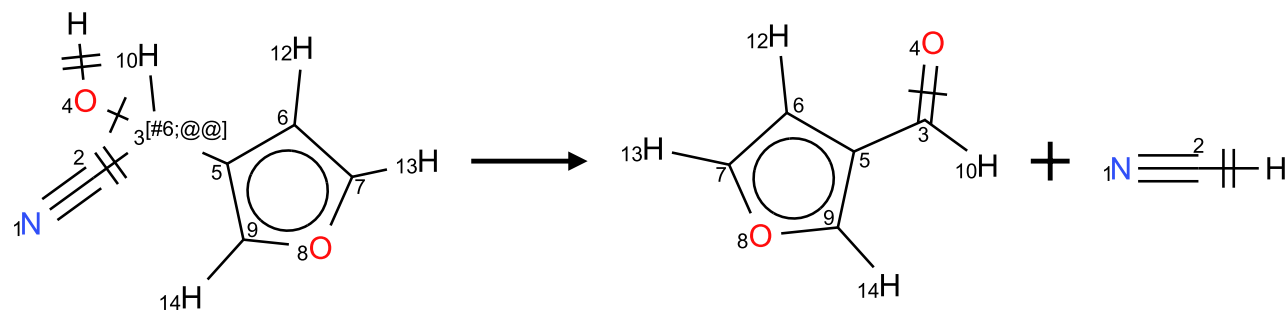
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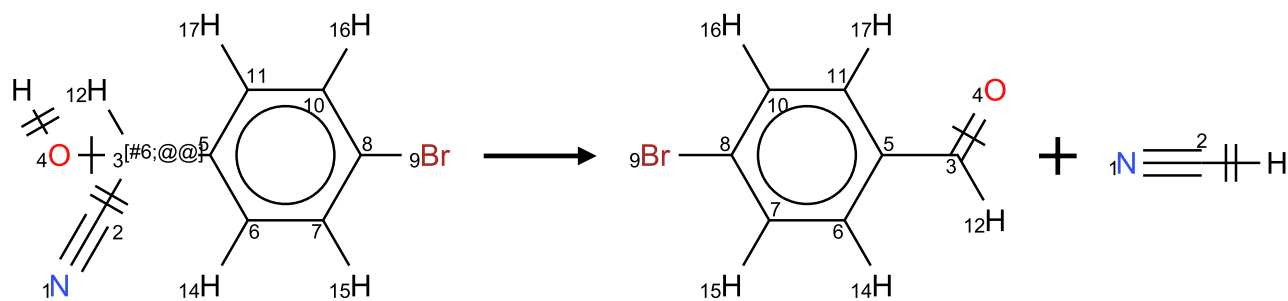
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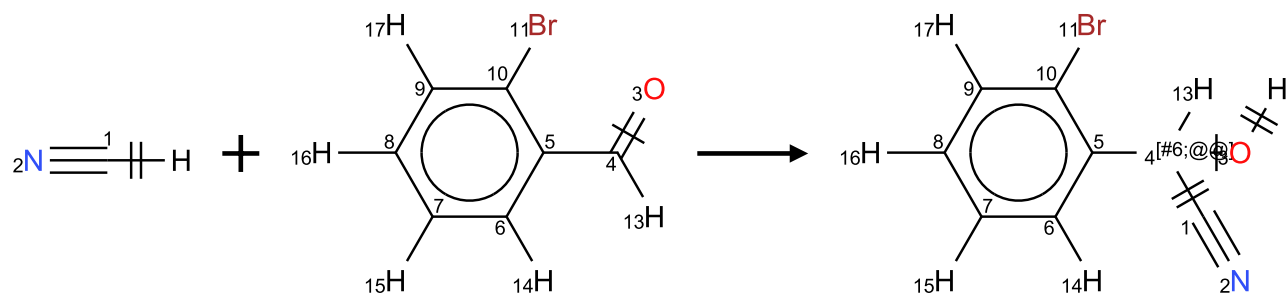
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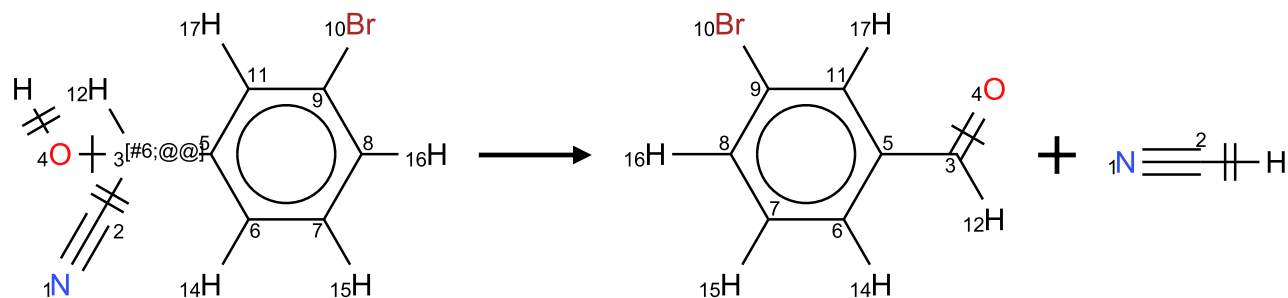
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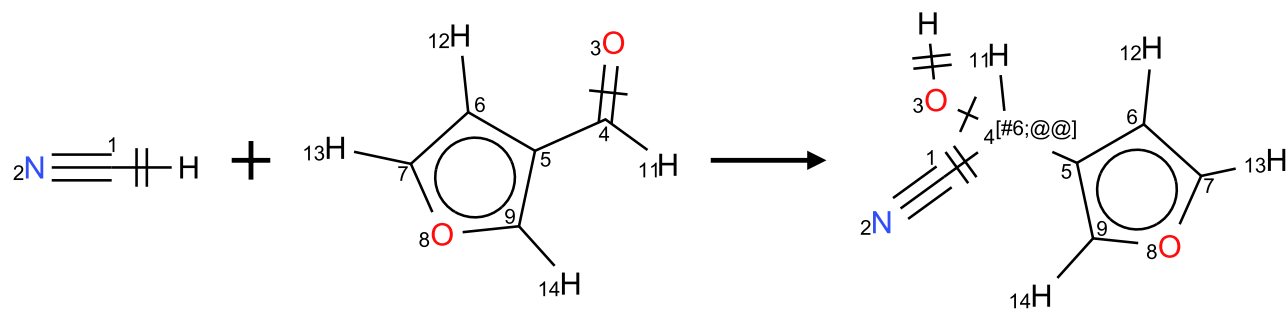
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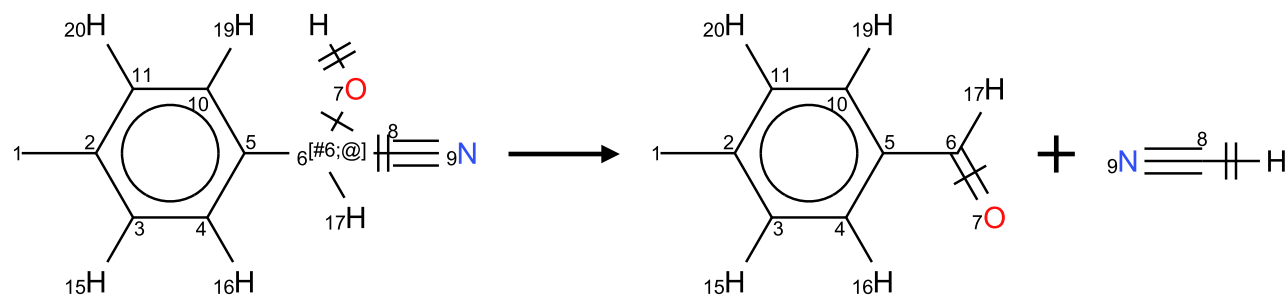
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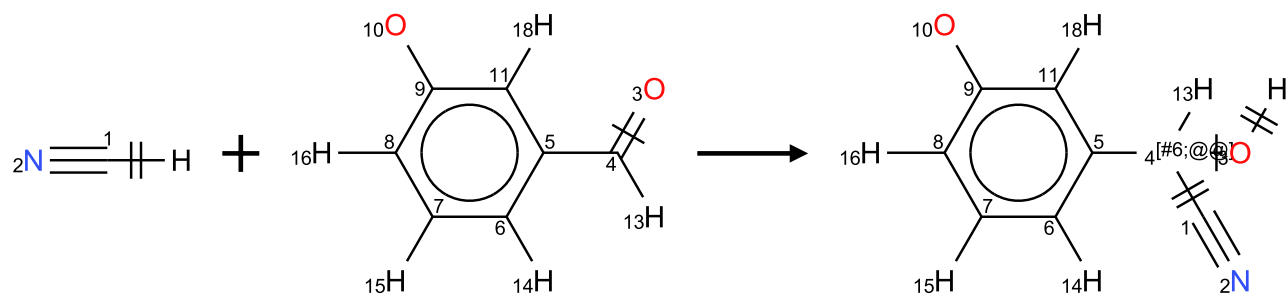
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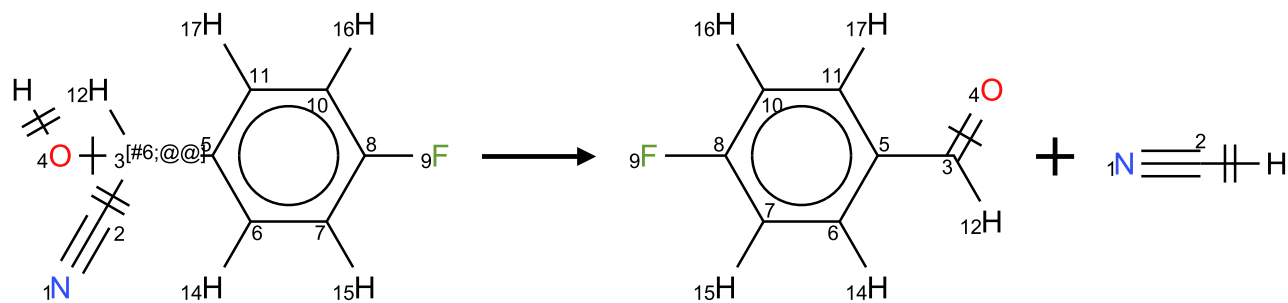
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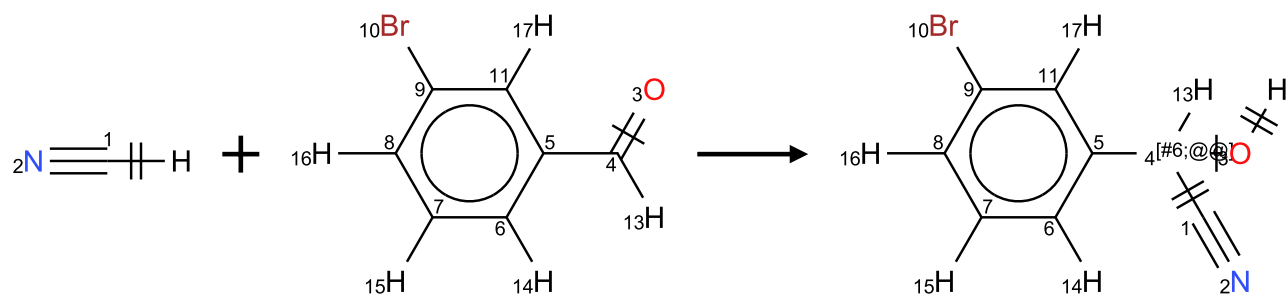
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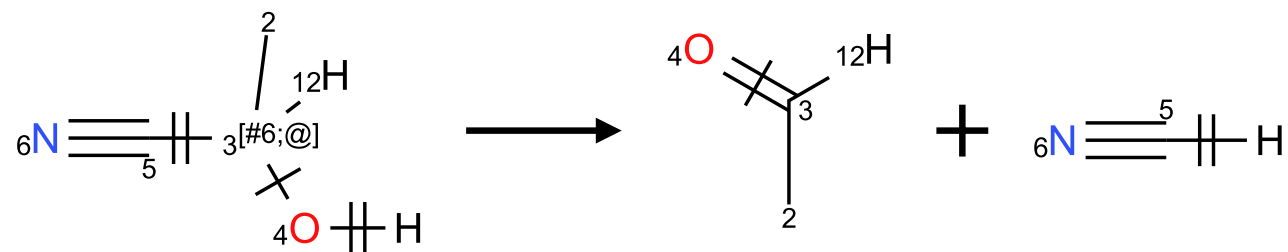
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(n=1)

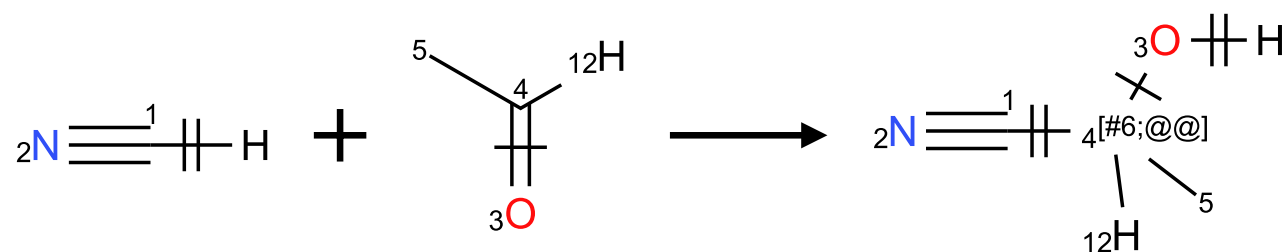


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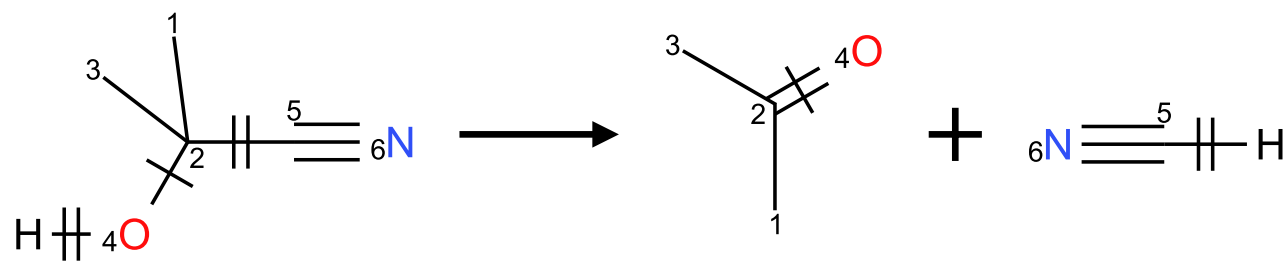


EVODEX-Cm (4 unique)

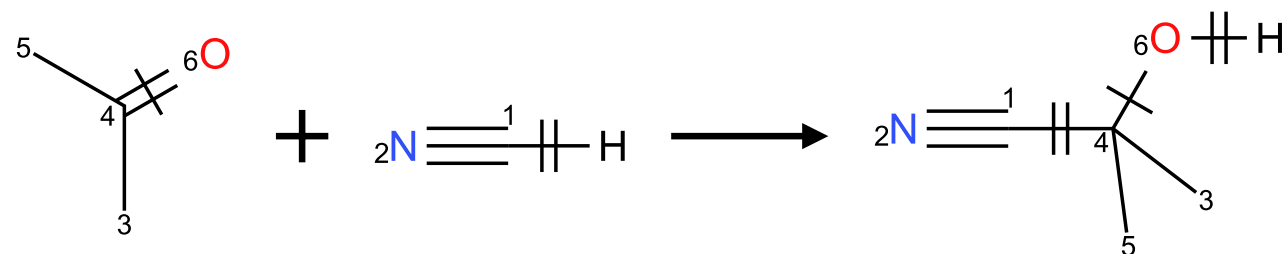
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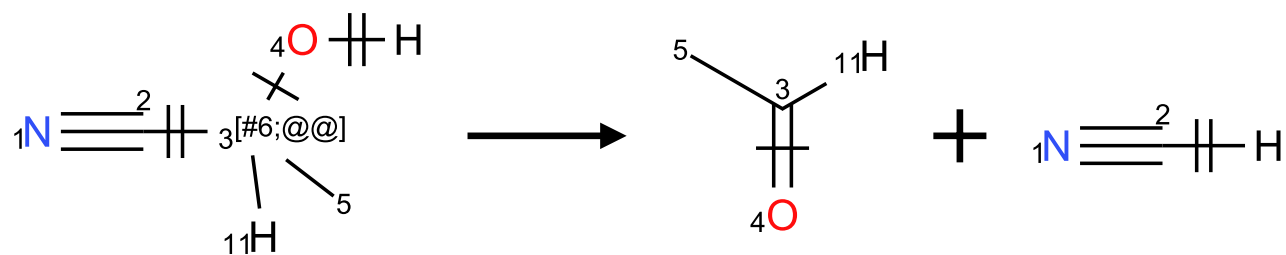
(n=45)



(n=24)



(n=24)



EVODEX-Bm (1 unique)

(n=138)

